Actual GF(2^83) norm-family research checkpoint

Date: 2026-10-17T02:59:11.657022+00:00

The cheap fixed-U transfer is refuted. All four actual relaxed families have dimension73, U-rank52, V-rank69 and joint rank73. Fixed-U slices have dimension21; enumerating2^52 slices is not admitted.

A missing transfer term was corrected: V=(h^2+h)e4+k(zs+z^2r+k*yT+z^3+h*z*yT). The earlier k=0=>V=0 claim is false. A GF4 literal-norm counterexample is preserved. Frozen generic support algorithms/timings are unchanged.

Repeated Q-root branch: the two eligible k=0 profiles gain69 linear constraints, leaving dimension4; all16 candidates per profile fail exact support. The two k=1 Q-roots are outside D. This closes only that norm branch at those four fixed profiles, not original roots or all outer parameters.

native-actual-norm-U-rank-n83-20261017-v1: passed. Actual norm-family affine U,V ranks reconstructed natively and checked against literal norms.

native-actual-repeated-Q-root-n83-20261017-v1: passed. Repeated Q-root branch intersected with actual relaxed families and all admitted support candidates checked.

native-actual-repeated-tangency-n83-20261017-v1: inconclusive. Fixed-h,k repeated tangency branch scanned through exact curve fibers and closed r,s formulas within its budget.

Tangency h=4208336d0d441df79c400,k=0: visited 126081/262144 source values; exact support count=None; complete=False.

Tangency h=4208336d0d441df79c400,k=1: visited 126081/262144 source values; exact support count=None; complete=False.

Tangency h=4208336d0d441df79c401,k=0: visited 126081/262144 source values; exact support count=None; complete=False.

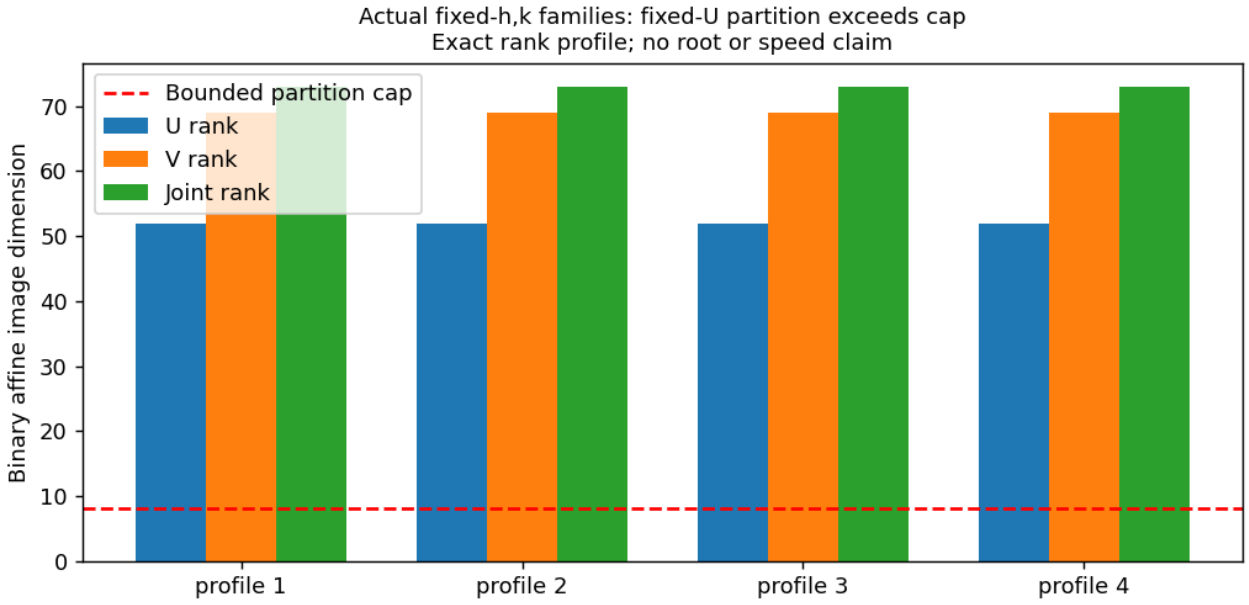
Tangency h=4208336d0d441df79c401,k=1: visited 126081/262144 source values; exact support count=None; complete=False.

Native independent controls:2304 literal norms,256 tiny family memberships,144 repeated-pair checks,15664 tangency parameter checks and28 Artin-Schreier right-hand sides. Each new native integration test passed. A native port omission of z^2*k was caught before freezing; its failed preparation log is preserved.

No qualified end-to-end gain, no fastest-external comparison, no matched hard F4/F5/F6 panel, no original source/auxiliary tuple completion, no novelty priority claim. Prime/odd adapter work remains separate.

Next: hoisted-denominator tangency variant; complementary squarefree/variable-U family reduction; exact odd prime interval adapter. See NEXT_QUEUE.md and REPRODUCTION.md.

Source PR2049 is updated. Reports remain local because automatic approval review rejected publication without explicit report-payload authorization.


Actual fixed-h,k families: fixed-U partition exceeds cap
Exact rank profile; no root or speed claim
Bounded partition cap
U rank
V rank
Joint rank
Binary affine image dimension
0
10
20
30
40
50
60
70
profile 1
profile 2
profile 3
profile 4